Mani Malek

Contact Information

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Code: `https://github.com/Anonymani?tab=stars`

Summary

Tech Lead and Research Scientist at Google DeepMind focusing on AGI readiness, with research interests at the intersection of AI safety, security, and interpretability. Concurrently exploring black-box and mechanistic interpretability to bridge the gap between observable model behaviors and their underlying neural mechanisms. Currently studying continual learning systems, with a particular focus on emerging misbehaviors.

Keywords and Highlights

- **AI Safety & Alignment**, **Interpretability (Black-box & Mechanistic)**, **Automated Red-Teaming**, **Post-Training (SFT, RL)**, **Model Steering**, **Agentic Systems**, **Continual Learning**, **Private Machine Learning**, **Differential Privacy**, **Federated Learning**, **NLP**, **Computer Vision**, **Recommendation Systems**, **Information Retrieval**, **Signal Processing**, **Image/Video Processing**, **Audio Processing**, **Data Processing**.
- Research Scientist with a solid record of delivering AI to production.
- Highly experienced in initiating and leading projects across multiple teams.
- Track record of mentoring junior and senior engineers in Industry and Academia.

Industry Experience

Google DeepMind

Tech Lead and Research Scientist — Nov. ’23 - present
Working on automated red-teaming, post-training (SFT, RL), black-box interpretability, mechanistic interpretability, model steering, agentic systems, and continual learning.

Meta (Facebook)

Technical Lead Manager, Reality Labs, Meta — Jan. ’23 - Nov. ’23
Leading private ML initiatives in AR/VR.
Setting future directions for private insights in metaverse.
Managing a small and elite team of scientists and engineers.

Lead Research Scientist in privacy in ML, Meta AI — Feb. ’20 - Dec. ’22
Initiated and led **Federated Learning** research in applied AI org, which resulted in multiple influential product launches.
Lead multiple **open sourcing** efforts within ML privacy.
Proposed and led various research efforts in **private ML**.

Software Engineer and Tech lead, Video Infra — Apr. ’17 - . Jan. ’20
Developed highly scalable **Audio/Video retrieval** algorithms.
Lead the research in video/audio understanding algorithms.

Magnitude Software (Simba Technologies), Vancouver, Canada — Dec. ’15 - . Mar. 17
Developed SQL language parsers and implemented various SQL APIs.
Developed efficient algorithms for **data processing at scale**.

Sasset Technology Inc., Vancouver, Canada — Jul. ’13 - . Nov. 15
Founded the company to improve EQ safety using computer vision.
Designed and developed the full software stack.
Managed all engineers.

Rigid Robotics Inc., Vancouver, Canada (Contract-student) — Jul. ’13 - Feb. ’14
Developed their **object tracking and segmentation** module.

Sidebuy Tech. Inc., Vancouver, Canada (Contract-student)

Developed their state-of-the-art **image search** module.
Developed an **integrated text/image retrieval** platform. Jul. '12 - Feb. '13

BroadbandTV Corp., Vancouver, Canada (Contract-student)
Developed an integrated **audio/video copy retrieval system**. Jul. '10 - Mar. '11

Education

The University of British Columbia, Vancouver, BC
- PhD, Electrical and Computer Engineering (GPA: 90) Sep. '07 - May '13
- Sub-specialization, Engineering Management Program (GPA: 91) Sep. '10 - Apr. '11

The University of Tehran, Tehran, Iran
- MSc, Electrical and Computer Engineering (GPA: 18 out of 20) Sep. '04 - Apr. '07
- BSc, Electrical and Computer Engineering (GPA: 17.5 out of 20) Sep. '00 - Sept. '04

Academic Experience

Professional Service & Activities
- Technical reviewer at NeurIPS. Jun. '20 - present
- Panelist on distributed computing and federated learning in IEEE CCNC Jan. '22.
- Representative of Facebook Industry Workshop in ICIP. Sep. '19
- Technical reviewer of various IEEE, Elsevier journals and conferences. Nov. '11 - Nov. '16
- Technical reviewer for IEEE T-IFS, IEEE J-STSP, Elsevier JVCI, IEEE MMSP, and IEEE ISCAS. Nov. '11 - Nov. '16
- Volunteer organizer of IEEE ICASSP 2013. May, '13
- Councilor at UBC Graduate Student Association. Mar. '08 - Oct. '08
- Member of the organizing team for UBC's Graduate Students Orientation. Aug. '07 - Sep. '07

Supervision, UBC, Vancouver, Canada
Co-supervised four undergraduate students at UBC through their final projects.
Mentored three undergraduate students who joined our research lab at UBC as interns. Jan. '10 - Aug. '11

Teaching Assistant, UBC, Vancouver, Canada
Served as teaching assistant for the following courses:
DSP systems, Introduction to Computation, **Signals & Systems**, Engineering Electromagnetic, and Electronic circuits. Sep. '08 - May. '12

Teaching Assistant, University of Tehran, Iran
Lab instructor and lecturer for **Electronic Circuits**. Jun. '03 - Sep. '04

Selected Honors and Awards

- IEEE Hector J. MacLeod Scholarship Award for student achievement. Mar. '13
- The University of British Columbia Ph.D. tuition fee award. Sep. '07
- University of Tehran, faculty of Engineering Award of excellence. Aug. '04
- Ranked in top 0.1% in different Iranian nationwide university exams.

Publications

- Visual prompt engineering for video models. *arXiv preprint arXiv:2607.25537, 2026.*
- The ACUTE Protocol: Operationalizing Language Model Activations for Better Calibration, Utility, and Trust. *International Conference on Machine Learning (ICML), 2026.*
- Hiding in Plain Text: Detecting Concealed Jailbreaks via Activation Disentanglement. *arXiv preprint arXiv:2602.19396, 2026.*
- Interpreting and Controlling Model Behavior via Constitutions for Atomic Concept Edits. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.*
- Concept Spaces in the Residual Stream of Diffusion Transformers. *CVPR Workshops, 2026.*
- ShieldGemma 2: Robust and Tractable Image Content Moderation. *arXiv preprint arXiv:2504.01081, 2025.*
- Joint Federated Learning and Personalization for on-Device ASR. *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), 2023.*
- Federated Ensemble Learning: Increasing the Capacity of Label Private Recommendation Systems. *IEEE Data Engineering Bulletin, 2023.*

- FEL: High Capacity Learning for Recommendation and Ranking via Federated Ensemble Learning. *arXiv preprint arXiv:2206.03852, 2022.*
- Federated Learning with Buffered Asynchronous Aggregation. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2022.*
- PAPAYA: Practical, Private, and Scalable Federated Learning. *Conference on Machine Learning and Systems (MLSys), 2022.*
- Antipodes of Label Differential Privacy: PATE and ALIBI. *Advances in Neural Information Processing Systems (NeurIPS), 2021.*
- Opacus: User-Friendly Differential Privacy Library in PyTorch. *arXiv preprint arXiv:2109.12298, 2021.*
- A local fingerprinting approach for audio copy detection. *Signal Processing, 2014.*
- Multimedia copy detection. *Ph.D. Thesis, University of British Columbia, 2013.*
- Image and Video Copy Detection Using Content-Based Fingerprinting. *Multimedia Image and Video Processing (Book Chapter), 2012.*
- A novel local audio fingerprinting algorithm. *IEEE International Workshop on Multimedia Signal Processing (MMSP), 2012.*
- A Fast Approximate Nearest Neighbor Search Algorithm in the Hamming Space. *IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI), 2012.*
- Fast matching for video/audio fingerprinting algorithms. *IEEE International Workshop on Information Forensics and Security (WIFS), 2011.*
- A Robust and Fast Video Copy Detection System Using Content-Based Fingerprinting. *IEEE Transactions on Information Forensics and Security, 2011.*
- Fine localization of photons on the pixellated detectors based on the analytical model. *Canadian Conference on Electrical and Computer Engineering (CCECE), 2010.*
- Robust video hashing based on temporally informative representative images. *International Conference on Consumer Electronics (ICCE), 2010.*
- Video Copy Detection Using Temporally Informative Representative Images. *IEEE International Conference on Machine Learning and Applications (ICMLA), 2009.*
- Topology Optimization for Backbone Wireless Mesh Networks. *Conference on Communication Networks and Services Research (CNSR), 2007.*